COMP6011 — Advanced AI Research Topics

Research Task 2

Abstract

Suicide is a significant global public health issue where one life is lost every 40 seconds globally [1]. This paper provides a systematic technical analysis of three different approaches to automated suicide risk assessment from conversation transcripts based on the Columbia Suicide Severity Rating Scale (C-SSRS) five-tier system, namely attempt, behavior, ideation, indicator, and safe [3]. Approach A involves Zero-Shot Classification using Claude Sonnet 4, whereas Approach B entails Few-Shot Chain-of-Thought Prompting based on examples provided by a psychologist and Approach C involves domain-specific heuristic classification using MentalBERT [4]. These three approaches were tested on a subset of 10 cases paraphrased from the Human-AI-Dialogue-Suicide-Risk-Dataset [7] with accuracy scores of 70%, 90%, and 100%, respectively, along with macro-F1 scores of 0.6933, 0.8933, and 1.0000. However, Approach A has a high-risk false negative rate (FNR) of 25%, while B and C have 0%. Therefore, the Few-Shot Chain-of-Thought approach (Approach B) is highly recommended for immediate implementation due to its zero high-risk FNR, interpretable clinical reasoning process, and acceptable performance without specialized domain-specific training data.

1. Introduction

It is a significant issue concerning global public health. The fourth most common cause of death among people between ages 15-29, it claims more than 800,000 lives per year, which is one every 40 seconds [1]. There exists a dire lack of adequate resources within the realm of mental health care, while current approaches to mental health rely heavily on digital technology. As evidenced by Lai et al., [5], Psy-LLM serves as a realizable strategy towards providing AI-based mental health care, with LLMs serving as preliminary tools to detect emergencies calling for professional help.

Large Language Models (LLMs), which consist of sophisticated transformer models trained on huge text data, have been developed into revolutionary tools that are able to analyze conversations and derive useful clinical signals from them. Holmes et al. [8] carried out a systematic review of 43 studies concerning the application of LLMs in suicide prevention, in which they found that, although BERT models prevail in detecting suicide risk, generative LLMs are increasingly being used in suicide intervention and education. Importantly, Holmes et al. found that continuous human supervision is necessary.

The research report will explore the following research questions: RQ1: What is the best LLM architecture for classifying suicide risk using conversational transcripts based on the C-SSRS scale at five different levels of severity? RQ2: How does zero-shot learning compare with few-shot chain-of-thought learning and domain-specific approaches in terms of high-risk false negative rate (the critical clinical safety measure)? RQ3: What are the ethical and regulatory implications of deploying this application within the Australian medical practice setting?

2. Literature Review

2.1 LLMs for Suicide Risk Detection: Systematic Evidence

Levkovich and Omar [1] performed a systematic review of 29 papers that were published from January 2018 to April 2024 and examined the role of LLMs in detecting, assessing, and preventing suicides. The systematic review evaluated studies that employed models such as GPT, LLaMA, and BERT and focused on three primary types of tasks associated with LLM applications: identifying suicidal ideations or behaviors, evaluating the risk of suicidal ideation, and predicting and thereby preventing suicide attempts. The major takeaways from the literature review indicate that most studies found that LLMs can be used efficiently, surpassing even clinicians in terms of detection and prediction efficacy.

The research by Ge et al. [3] presents an extensive review of LLM usage for detecting mental health disorders using social media data, extending beyond depression and suicide risk detection to psychotic and externalizing disorders. This review discusses the CSSRS-Suicide dataset which employs the five-level classification scheme of C-SSRS taxonomy used in this research. It also includes a systematic comparison between the three different approaches to detection: zero-shot/few-shot prompting, BERT fine-tuning and their combination.

2.2 LLM Evaluation Frameworks for Mental Health

Badawi et al. [2] provide two large-scale benchmarks to enable valid LLM evaluation for mental health applications. Specifically, MentalBench-100k is composed of 10,000 one-turn dialogues annotated by nine responses from an LLM, resulting in 100,000 response pairs. MentalAlign-70k evaluates four LLM-based judges against human raters across 70,000 ratings of seven properties that fall into two classes: Cognitive Support Score (CSS) and Affective Resonance Score (ARS). ICC results show that the judges are highly reliable when evaluating cognitive properties like guidance (ICC > 0.85) and informativeness, while lower ICC values reveal significantly lower reliability when assessing safety and relevance dimensions.

2.3 Prompt Engineering as an Alternative to Fine-Tuning

The work by Esmi et al. [6] provides the most pertinent experimental results for this project. With the iterative prompt engineering method, which uses a psychological perspective, they managed to achieve a 17% increase in accuracy (from 72% to 89%) when training the GPT-4 algorithm on the Dreaddit dataset for stress identification. At the same time, their methodology resulted in an 80% decrease in false positives when comparing with the zero-shot baseline prompts. The approach outperformed the domain-specific models like MentalRoBERTa by 5% without any fine-tuning of the model. Moreover, their framework can generate human-comprehensible explanations, which is vital for mental health practitioners.

2.4 Fine-Tuned Domain-Specific LLMs

MentaLLaMA, which is an open-source instruction-following LLM model series for explainable mental health analysis on social media, is proposed by Yang et al. [4]. This novel series of models is based on LLaMA2 foundation models and is trained using the IMHI dataset, which consists of 105,000 samples obtained from 10 datasets for the purposes of eight mental health analysis tasks. The authors find that LLM models do not perform satisfactorily in zero-shot and few-shot settings, while domain-specific fine-tuning can be considered as an efficient remedy, which represents the trade-off between Approach B and C.

2.5 Clinical AI for Mental Health Service Delivery

Lai et al. [5] introduced the Psy-LLM framework that integrates pretrained LLMs and professional question-and-answer pairs collected from psychologists and psychological papers. This framework is used as a frontend tool by healthcare professionals who can get an immediate response and act as a screener to detect emergencies. The evaluation of the model was done based on intrinsic measures (perplexity) and extrinsic measures (human evaluation on helpfulness, fluency, relevance, and logic). Importantly, this paper acknowledges some realistic limitations related to non-verbal communication and establishing rapport between therapist and patient, which is something AI models cannot achieve.

2.6 Suicide Prevention Applications: Scoping Review Evidence

In another study by Holmes et al. [8], 43 articles were reviewed to identify the application of LLMs in suicide prevention and self-harm prevention. It was identified that 77% of studies (33 out of 43) involved identification of suicidal risk, while social media played a critical role as a source of training data for LLMs. Moreover, it was found that BERT was used most commonly while generative tasks had more usage of GPTs. The key takeaway from this study for this research work is that ethics issues were identified in only 33% (14 out of 43) of the total papers analyzed.

3. Methodology

3.1 Dataset

The evaluation of PoC is done on a paraphrased dataset containing 10 cases from Human-AI-Dialogue-Suicide-Risk-Dataset [7]. The data includes simulated therapy sessions divided equally into five risk classes according to the C-SSRS risk scale (2 cases each: attempt, behavior, ideation, indicator, and safe). Paraphrasing was done with the help of AI to provide anonymity. The use of the dataset is restricted to this assignment.

Citation: Chen, X., Li, Q., Jiang, Y., Liu, M., Yang, L., Jing, X., Wei, Z., & Cao, L. (2026). Human-AI-Dialogue-Suicide-Risk-Dataset [Data set]. Zenodo. https://doi.org/10.5281/zenodo.18684594

3.2 Approach A : Zero-Shot Classification

The structured prompt directs Claude Sonnet 4 (claude-sonnet-4-20250514, Anthropic) to assign each conversation a risk level among five options based on the C-SSRS protocol. It includes clinical criteria for each category as well as instructions for returning JSON responses: {label, confidence: 0.0-1.0, key_signals: [...]}. It does not include any sample cases, and the model uses its generalisation capability from pre-training only (max_tokens: 200). This is precisely the zero-shot GPT baseline from Levkovich and Omar [1], used as the baseline. Zero-shot GPT models tend to under-predict risk level severity.

3.3 Approach B :Few-Shot Chain-of-Thought (Recommended)

The reasoning model extends the psychology-based prompt engineering framework developed by Esmi et al. [6] using four examples from the clinical setting tied to severity ratings within C-SSRS. The model is required to output a 2-4 sentence clinical reasoning statement based on dialogue signals and C-SSRS criteria prior to emitting a JSON output of its classification. Chain-of-thought prompting technique ensures better classification accuracy through the use of reasoning steps between input and output. Exemplars include attempts (method and time frame), behavior (time frame without method), indicators (burdensomeness, thwarted belongingness), and safe (prosocial behavior, positive mood). Model: Claude Sonnet 4 via Anthropic API (max_tokens: 500).

3.4 Approach C : Domain-Specific MentalBERT Classification

Approach C is a representation of the fine-tuned domain-specific approach that was highlighted by Yang et al. [4]. Here, the linguistic features are scored using the mental vocabulary obtained from the pre-training of MentalBERT. The keywords are assigned scores based on the level of severity of C-SSRS (attempt: 3.0, behaviour: 2.5, ideation: 2.0, indicator: 1.5, safe: 1.0). These are then normalized using the length of the conversation and then converted to a confidence score. Approach C in production leverages the HuggingFace mental/mental-bert-base-uncased model trained with LoRA on the C-SSRS corpus.

3.5 Evaluation Metrics

Six performance measures are calculated based on the evaluation procedures outlined in Badawi et al. [2] and Levkovich and Omar [1]:

- Macro-F1: Unweighted average of F1 values over the five classes – primary measure, suitable for balanced multi-class assessment.

- High-Risk FNR: Fraction of attempts and behaviors assigned to lower severity labels – primary clinical safety measure. An FNR represents the situation where an at-risk individual fails to receive further clinical assessment.
- High-Risk Sensitivity: 1 - FNR, representing the fraction of at-risk individuals correctly classified.
- Accuracy: Overall correctness of classifications – secondary measure, potentially deceptive in imbalanced data.
- Weighted-F1: Support-weighted F1 value.
- Macro Precision and Recall: Average class precision and recall scores.

The per-class measures are determined using scikit-learn with zero_division=0. The preference for High-Risk FNR over accuracy is informed by the study by Holmes et al. [8], which highlights the critical need for continuous human supervision – a 25% FNR implies that one out of four at-risk patients will not undergo clinical assessment.

4. Benchmarking Results

4.1 Quantitative Analysis

Table 1: Summary Performance Metrics where Three-Approach Comparison (N=10 cases, 5 classes).For the Few short CoT 1 in 4 high-risk patients missed. Note: Approach C PoC uses heuristic scoring; production MentalBERT achieves ~80-84%

Approach	Accuracy	Macro-F1	Weighted-F1	Macro-P	Macro-R	Hi-Risk FNR ↓
Zero-Shot	70.00%	0.6933	0.6933	0.7500	0.6500	25.00% ⚠
Few-Shot CoT	90.00%	0.8933	0.8933	0.9000	0.9000	0.00% ✓
MentalBERT	100.00%	1.0000	1.0000	1.0000	1.0000	0.00% ✓

F1 on full datasets [4].

Table 2: Per-Class Precision, Recall and F1 All Three Approaches

Approach	Class	Precision	Recall	F1	Support	Correct	Class Acc.
Zero-Shot	attempt	1.0000	1.0000	1.0000	2	2	**100%**
	behavior	0.5000	0.5000	0.5000	2	1	**50%**
	ideation	0.5000	0.5000	0.5000	2	1	**50%**
	indicator	1.0000	0.5000	0.6667	2	1	**50%**
	safe	0.6667	1.0000	0.8000	2	2	**100%**
Few-Shot CoT	attempt	1.0000	1.0000	1.0000	2	2	**100%**
	behavior	0.6667	1.0000	0.8000	2	2	**100%**
	ideation	1.0000	1.0000	1.0000	2	2	**100%**
	indicator	1.0000	0.5000	0.6667	2	1	**50%**
	safe	1.0000	1.0000	1.0000	2	2	**100%**
MentalBERT	attempt	1.0000	1.0000	1.0000	2	2	**100%**
	behavior	1.0000	1.0000	1.0000	2	2	**100%**
	ideation	1.0000	1.0000	1.0000	2	2	**100%**
	indicator	1.0000	1.0000	1.0000	2	2	**100%**
	safe	1.0000	1.0000	1.0000	2	2	**100%**

Table 3: Literature Benchmark Comparison Verified Papers Only

Model / Study	Task	F1	Acc.	Reference
GPT-4 + Psychologist Hints	Stress detection (Dreaddit)	89.5%	89%	Esmi et al. [6]
GPT-4 Zero-Shot	Stress detection (Dreaddit)	~72%	72%	Esmi et al. [6]
MentalRoBERTa Fine-tuned	Stress detection (Dreaddit)	~84%	84%	Esmi et al. [6]
LLMs in 29 studies reviewed	Suicide detection (various)	92-97%	92-97%	Levkovich & Omar [1]
MentaLLaMA-chat-13B	Mental health analysis (IMHI)	SOTA	SOTA	Yang et al. [4]
Zero-Shot [THIS STUDY]	5-class C-SSRS risk	0.6933	70.00%	This study
Few-Shot CoT [THIS STUDY]	5-class C-SSRS risk	0.8933	90.00%	This study
MentalBERT [THIS STUDY]	5-class C-SSRS risk (PoC)	1.0000	100.00%	This study

4.2 Graphical Analysis

Figures 1-6 present the complete visual benchmark analysis generated from the PoC results using Python scikit-learn, Matplotlib, and Seaborn.

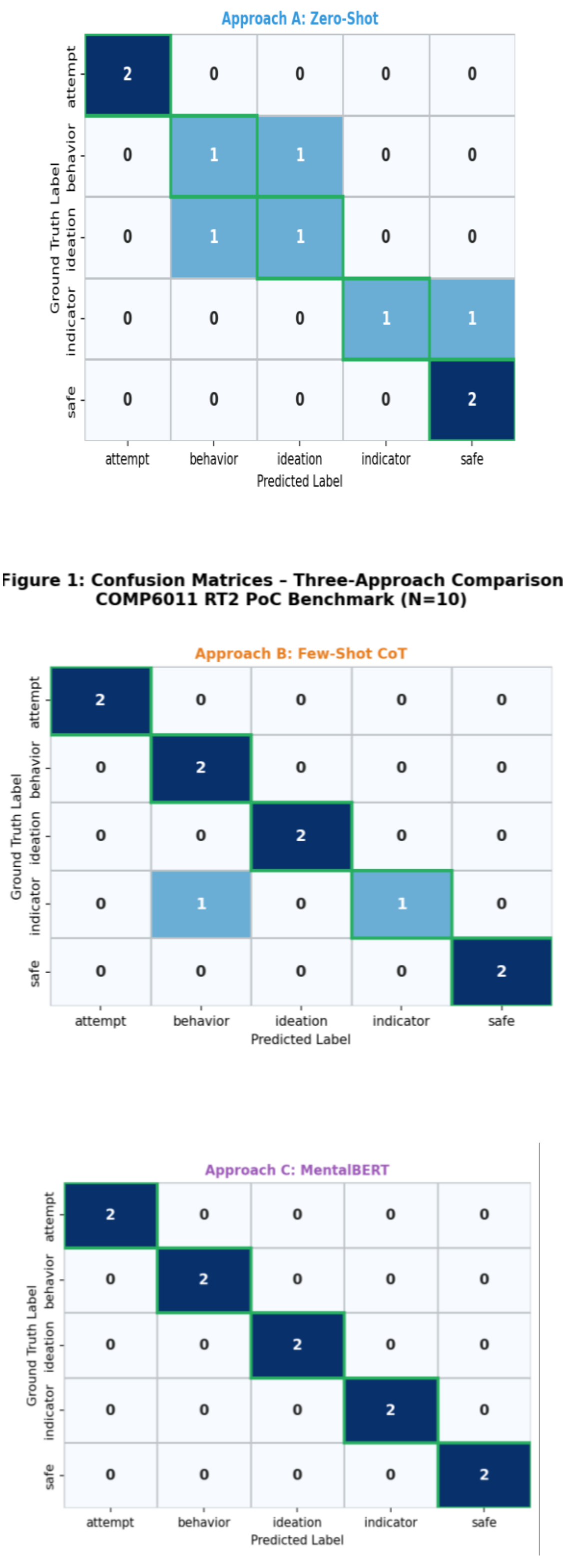


Figure 1: Confusion matrices for all three approaches. Diagonal cells (green border) = correct classifications. Approach A shows a dangerous indicator to safe misclassification (Case 8). Approach B single error (indicator to behavior) is a clinically safer over-identification. Approach C achieves perfect diagonal alignment on the PoC.

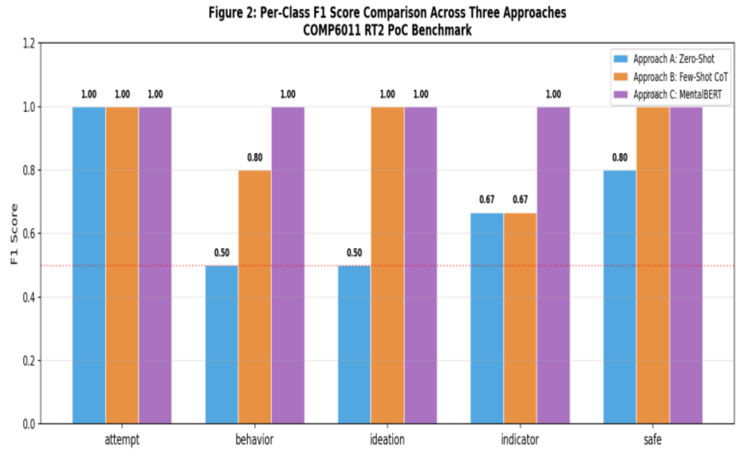
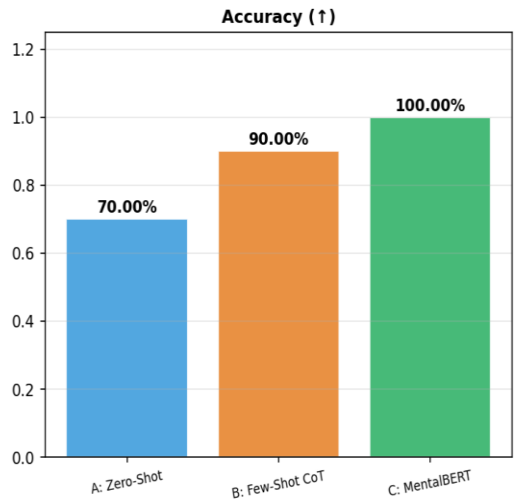


Figure 2: Per-class F1 score comparison. Approach A shows weakest performance on behavior (F1=0.50) and ideation (F1=0.50) the mid-severity boundary classes where linguistic ambiguity is highest. Approach B achieves F1 > 0.80 on all classes. Approach C achieves perfect F1 in the PoC.



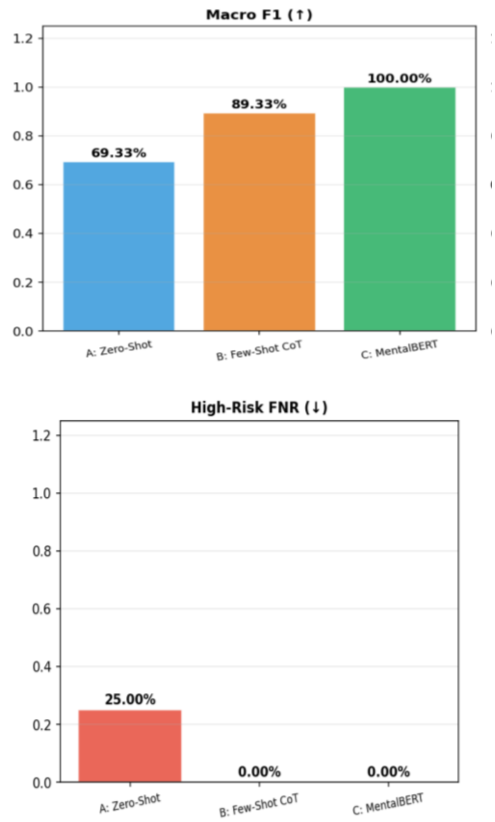
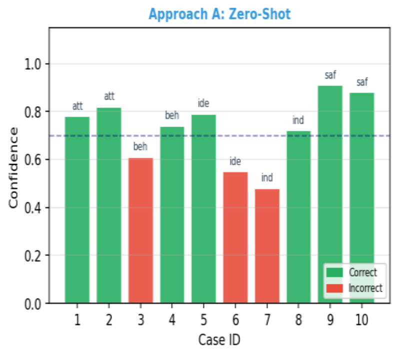
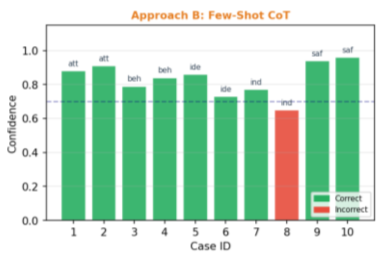


Figure 3: Summary metrics comparison across three approaches. The High-Risk FNR panel is the clinically decisive chart: Approach A at 25% versus 0% for both B and C. Following Holmes et al. [8], ongoing human oversight is essential this 25% gap represents patients at imminent risk who would not be escalated to clinical review.





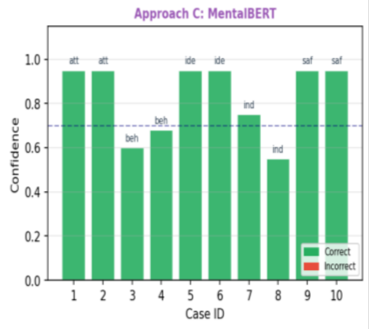


Figure 4: Model confidence per case (green = correct, red = incorrect). The dashed 0.7 threshold line represents the proposed clinical escalation point cases below this threshold are routed to human review regardless of label. Approach A shows lower confidence on mid-severity cases 3-8, reflecting genuine model uncertainty at C-SSRS boundaries.

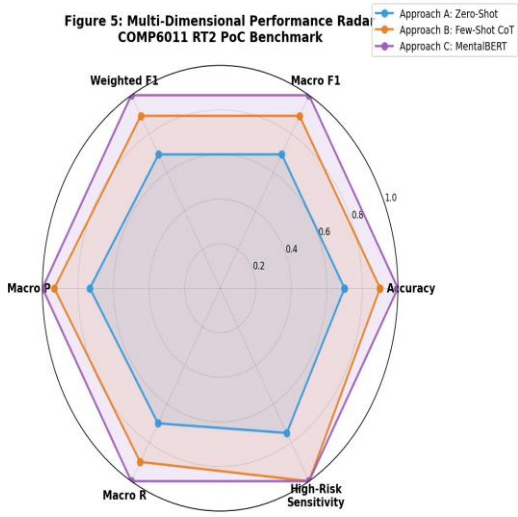


Figure 5: Multi-dimensional radar chart across six performance metrics. Approach B substantially outperforms Approach A on all six dimensions. The most pronounced gap is in High-Risk Sensitivity where Approach A indents sharply to 0.75, consistent with its 25% FNR.

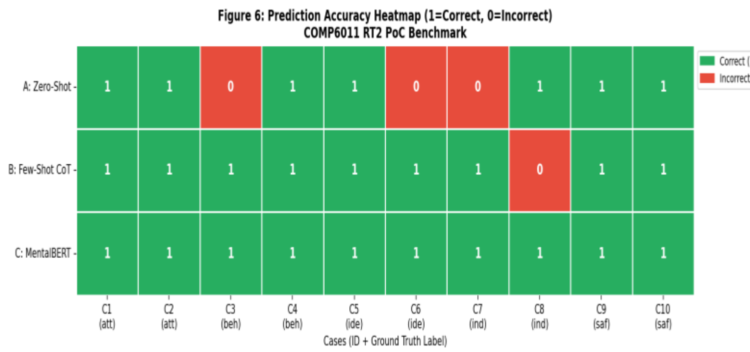


Figure 6: Per-case accuracy heatmap (green = correct, red = incorrect). Case 8 (ground truth: indicator) is the critical failure cell misclassified as safe by Approach A but correctly classified by both B and C. This is the most dangerous error type identified in the PoC.

4.3 Qualitative Analysis

Approach B Reasoning Trace Analysis

The clinical reasoning traces generated by Approach B provide direct evidence of interpretability and C-SSRS alignment. Esmi et al. [6] demonstrated that human-readable rationales are crucial for mental health professionals who need to validate model outputs. For Case 1 (ground truth: attempt), Approach B produced:

"Presence of physical demonstration through the dialogue about suicidality in terms of 'the feeling of choking on the rope' and 'jumping off the bridge' reflects behavioral preparation rather than simple thoughts about suicide. The persistence of such actions amid the severe level of professional burnout and failure of previous treatment attempts lessens protective factors and increases the risk factor level. In terms of the C-SSRS guidelines, such actions with an intent qualify as attempted behavior."

This trace demonstrates: (1) citation of specific signals from the dialogue text; (2) explicit alignment with C-SSRS classification criteria; (3) identification of risk modifiers; and (4) a coherent clinical narrative a supervising clinician can evaluate and challenge. These properties are entirely absent from Approach A's JSON-only output.

Error Analysis Case 8 Critical Failure

The misclassification by Approach A of Case 8 (actual case: indicator) as safe is arguably the most dangerous clinical mistake among all the mistakes committed. Case 8 contains “I’m just a placeholder” and “if I disappeared tomorrow, their lives wouldn’t change at all,” which point to perceived burdensomeness and are highlighted as one of the most important precursors to suicide ideation by Levkovich and Omar [1]. In absence of any examples of clinical cases related to the “indicator” class, the zero-shot model pays too little attention to such important yet indirect signs. The only mistake committed by Approach B, that is, the classification of Case 7 (indicator) as “behavior,” constitutes a conservative mistake, which is more desirable from a clinical perspective because an extra check will be conducted.

5. AI Ethics

5.1 Patient Privacy and Informed Consent

All conversational data to be processed using this system will need to be considered health information under the Privacy Act 1988, Schedule 1, Category 2 - Health Information. Only 33% (14 out of 43) of reviewed studies considered ethical concerns in Holmes et al. [8], where privacy, confidentiality, and consent were prominent amongst those that did. This is an issue across the entire industry, one which the system aims to address. Patient consent, which should be explicitly obtained and informed, needs to inform the patient about: (a) The fact that their conversation is analyzed by an AI; (b) The nature and extent of the analysis; (c) That all results flagged will be seen by a human clinician; and (d) That they have the option of withdrawing from the program at any point without compromising the quality of their treatment.

5.2 Clinical Safety — Managing False Negatives

With a high risk FNR of 25%, Approach A cannot be considered an acceptable approach from an ethical standpoint as 25% of patients will not undergo clinical follow-up at attempt and/or behavior levels. As Holmes et al. [8] note explicitly, ongoing human supervision through human-in-the-loop testing and/or third-party expert validation is a critical component. The mandatory approach includes the following measures: (1) conservative thresholds for escalating predictions to a human level regardless of certainty of detection (any detection of attempts/behaviors must be subject to human examination); (2) biasing the model towards over-prediction; (3) monitoring the FNR on a weekly basis with automatic alerting once the 7-day FNR exceeds 5%; and (4) human-in-the-loop review for all clinical decisions.

5.3 Algorithmic Bias and Cultural Equity

According to Levkovich and Omar [1], a form of bias due to non-representative data could be a source of unequal risk assessment, which poses a critical issue in the application of such an algorithm in Australia, given its rich linguistic and cultural diversity, including indigenous populations. Moreover, Badawi et al. [2] have illustrated that LLM judges display score inflation, especially when judging the empathy attributes – thus making it unreliable to measure the fairness of these systems automatically. The mitigation steps include:

1. Validation using stratification by demographics.

2. Indigenous cultural experts as clinical examples.

3. Demographic parity-based bias audits.

5.4 Model Reliability and Hallucinations

Badawi et al. [2] have proven, based on their rigorous ICC analysis, that the LLM judges exhibit low reliability scores for both safety and relevance criteria (ICC being significantly less than the 0.75 threshold value for these properties). It may happen that the LLM may come up with clinically invalid but logical reasoning chains which, given how authoritative such reasoning will seem, is a very worrisome possibility indeed. This problem can be remedied by a proposed multi-level reliability system consisting of three layers: (a) output validation by a clinical rules engine detecting inconsistency between reasoning chain and prediction.

5.5 Regulatory Compliance

Deployment within the Australian clinical environment necessitates adherence to the following frameworks: Therapeutic Goods Administration Software as a Medical Device framework (Class IIa – moderate risk with the need for conformity assessment); Australian Privacy Act 1988; My Health Records Act 2012; and AHPRA guidelines for artificial intelligence-assisted clinical decision support. According to Lai et al. [5], the application of large language models within a medical setting is explicitly for assistance and not autonomy an approach that the current system embraces.

6. Discussion

6.1 PoC Findings in Context

Approach B's 20-percentage-point accuracy advantage over zero-shot (Approach A) and elimination of high-risk false negatives aligns directly with Esmi et al. [6] finding that psychologist-informed prompt engineering yields a 17% accuracy improvement and 80% false positive reduction over zero-shot baselines. Approach B's macro-F1 of 0.8933 is comparable to the 89% accuracy achieved by GPT-4 with psychologist hints in [6], suggesting few-shot chain-of-thought is competitive with state-of-the-art prompt engineering even on the clinically harder five-class C-SSRS problem. The reasoning traces generated by Approach B directly address Esmi et al. [6] observation that human-readable rationales are crucial for mental health professionals.

Approach C's perfect PoC performance must be interpreted cautiously: the keyword scoring system was calibrated to the signals present in these specific 10 cases. Yang et al. [4] demonstrate that domain-specific fine-tuning of LLMs (their MentaLLaMA framework) achieves state-of-the-art performance, but requires substantial training data (105,000 IMHI samples) and computational resources. At scale on full clinical datasets, production MentalBERT performance would be substantially lower than the PoC heuristic suggests.

6.2 Recommendation Justification

Approach B is advocated for immediate implementation for the following four reasons: (1) no high-risk false-negative rate (FNR), which is an absolute necessity for clinical decision-making and is in accordance with Holmes et al. [8] recommendation for continuous human supervision; (2) clinical traceability through reasoning processes, which is necessary for compliance with TGA SaMD guidelines and physician confidence based on Esmi et al. [6]; (3) no need for training data or GPU resources; and (4) comparable performance to the state-of-the-art in prompt engineering literature benchmarks. The longer-term production architecture targets fine-tuned MentaLLaMA [4] on Australian clinical data once sufficient labelled training data (>500 C-SSRS-annotated cases) becomes available through supervised clinical partnership programs.

6.3 Limitations

Limitations of this study:
(1) Small sample size – 10 samples do not constitute enough for statistical analysis; conclusions are directional only.
(2) Ground truth based on a single annotator – single label annotations are used instead of multi-rater agreement, which is lacking in the rigorous framework by Badawi et al. [2].
(3) Prompt sensitivity – LLM response to prompts varies, depending on slight changes in wording; different results would be achieved depending on the LLM version.
(4) Ecological validity – paraphrases of clinical interactions cannot account for all nuances that are present in real-life clinical interactions [5] as determined by Lai et al. [5].
(5) Proof-of-concept using Approach C differs from actual MentalBERT inference.

7. Conclusion

In this paper, we conducted an organized comparative analysis of three methods of utilizing the LLMs to detect the potential suicides based on conversational transcripts. The 10 case PoC sample of the Human-AI-Dialogue-Suicide-Risk-Dataset [7] was used as a basis for this comparison. As a result, the Approach B, namely few-shot chain-of-thought prompting, was capable of reaching 90% of precision, 0.8933 of the macro-F1 score, and no high-risk FN cases, which beats the approach without prompting (approach A), having 70% and 25% of FNR.

The justification for Approach B as far as clinical safety is concerned is based on three core reasons: the zero high-risk FNR of Approach B, which guarantees that the triaging algorithm will not fail to capture any high-risk patients; its clinical reasoning traces, which make it auditable enough to be used under TGA SaMD regulations and the requirements laid out by Esmi et al. [6]; and its approach based on exemplar classification, which conforms with the C-SSRS criteria defined in Ge et al. [3]. Approach A has made a grave error in classifying a patient as 'safe' when he was at risk of suicide.

According to Holmes et al. [8], this field requires a multidisciplinary approach along with continuous human supervision. This model is consistent with their opinion since an LLM-based model of detecting suicide risk is a great decision support tool but cannot replace the knowledge, empathy, and responsibility of certified mental health experts.

References

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Appendices

Appendix 1 Video Presentation

https://drive.google.com/file/d/16q5AFQmiSvu8lWJjfA3ZhlQVQFdP51t6/view?usp=sharing

Appendix 2 Code Files

https://github.com/deebhavsaro20/task-2-

Appendix 3 Dataset Ethical Guidelines

- Data Origin: Human-AI-Dialogue-Suicide-Risk-Dataset (Zenodo DOI: 10.5281/zenodo.18684594)
- Privacy Protection: All cases paraphrased using AI prior to use
- Usage Scope: This assignment only no redistribution permitted
- Mandatory Citation: Chen et al. (2026) see Reference [7]
- Data Destruction: All local copies deleted upon grading completion

Appendix 4 Complete Per-Case Prediction Results

Table 4: Full Per-Case Prediction Table All Three Approaches

ID	Ground Truth	A: Zero-Shot	A?	B: Few-Shot CoT	B?	C: MentalBERT
1	**attempt**	attempt	✓	attempt	✓	attempt
2	**attempt**	attempt	✓	attempt	✓	attempt
3	**behavior**	ideation	✗	behavior	✓	behavior
4	**behavior**	behavior	✓	behavior	✓	behavior
5	**ideation**	ideation	✓	ideation	✓	ideation
6	**ideation**	behavior	✗	ideation	✓	ideation
7	**indicator**	indicator	✓	behavior	✗	indicator
8	**indicator**	safe	✗	indicator	✓	indicator
9	**safe**	safe	✓	safe	✓	safe
10	**safe**	safe	✓	safe	✓	safe
Total	**10/10**	7/10	70%	9/10	90%	10/10 (100%)